

Problem Solving Practice

HW Problem #4

CSE4152
Sogang University



Half-Circle Property

There is an circle C centered at $(0,0)$ and there are many points on the circumference (원주) of the circle. Let us denote the set of points by S . If there exists a line L passing through point $(0,0)$ such that all points of S lie on the one side of L , then we say that S satisfies 'Half-circle property'. For a given set of point, design an algorithm that finds whether the set satisfies half-circle property or not. And compute its time and space complexity.

Example:

1st example: $(0.5, 1.0), (1.0, -0.5), (0.5, 1.0)$.

2nd example: $(0.5, 1.0), (1.0, -0.5), (-0.5, -1.0), (-1.0, 0.5)$.

Solution of Examples:

1st example point set satisfies half-circle property.

2nd example point set does not.

K Fire Station

Sogang City consists of **N buildings and $N - 1$ roads**, forming a tree structure. (In other words, there is exactly one path between any two buildings, and all buildings are connected.) The roads are bidirectional, and each road has its own length.

After being invaded by neighboring Sinchon cities harboring hostility toward Sogang, all of Sogang City's fire stations were destroyed. Therefore, you are tasked with rebuilding **K fire stations** in Sogang City (where $K = 1$ or $K = 2$). The goal is to place the fire stations in optimal locations so as to minimize the maximum distance from any building to its nearest fire station.

Fire stations can be placed only on roads, and they may also be placed at the endpoints of the roads. (In other words, they can be built on top of existing buildings.)

K Fire Station (Guide Questions)

1. When a single fire station is given, present the most naive method of solving the problem.
2. If, from an arbitrary vertex S you find the farthest vertex A , and then from A you again find the farthest vertex B , what does the resulting path $P(A, B)$ represent? Provide a logical explanation of this.
3. If you place the fire station at A or B , what is the maximum distance from any building to the fire station?
4. When only one fire station can be placed, present an algorithm to solve the problem and prove its correctness.
5. Now suppose you want to place two fire stations. In this case, divide the tree into two regions by choosing an edge, and place a fire station optimally in each subtree. Which edge should you choose in order to reduce the maximum distance compared to placing only one fire station?
6. Based on the above questions, present an algorithm to solve the problem and prove its correctness.

K Fire Station (Example)

